Uday Deshpande

FPGA Design Engineer

RESUME_1252

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Total Experience: 10 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 10 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: FPGA, ILA, Intel Quartus, Python, Verilog, Xilinx Vivado, Timing Closure, AXI4, DSP, Tcl, UART, VHDL

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

Lead FPGA Design Engineer | Ridgeview Engineering | Jan 2023 - Present

- Improved maximum operating frequency by 14% after timing-path optimization
- Reduced FPGA resource utilization by 38% through arithmetic sharing
- Lowered end-to-end data-path latency by 12% using pipelined logic
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

FPGA Design Engineer | JuniperWorks Systems | Jan 2019 - Dec 2022

- Lowered end-to-end data-path latency by 22% using pipelined logic
- Verified 358 protocol corner cases with repeatable simulation suites

Associate FPGA Design Engineer | Asterline Engineering | Jul 2016 - Dec 2018

- Improved maximum operating frequency by 15% after timing-path optimization
- Reduced FPGA resource utilization by 29% through arithmetic sharing

PROJECTS

Camera Data Bridge | Tcl, UART, Python, AXI4

- Designed AXI streaming logic, frame buffers and clock-domain boundaries.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

Protocol Adapter Core | Verilog, Intel Quartus, DSP, VHDL

- Developed reusable SPI and UART modules with loopback tests.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Northbridge College of Technology, 2016
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

FPGA Architecture Workshop (2019)